实验记录_20260227_通过裁剪奇点来验证双隐藏层中不同奇点权重系数的重要性

实验名称：通过裁剪奇点来验证双隐藏层中不同奇点权重系数的重要性

实验时间：20260227

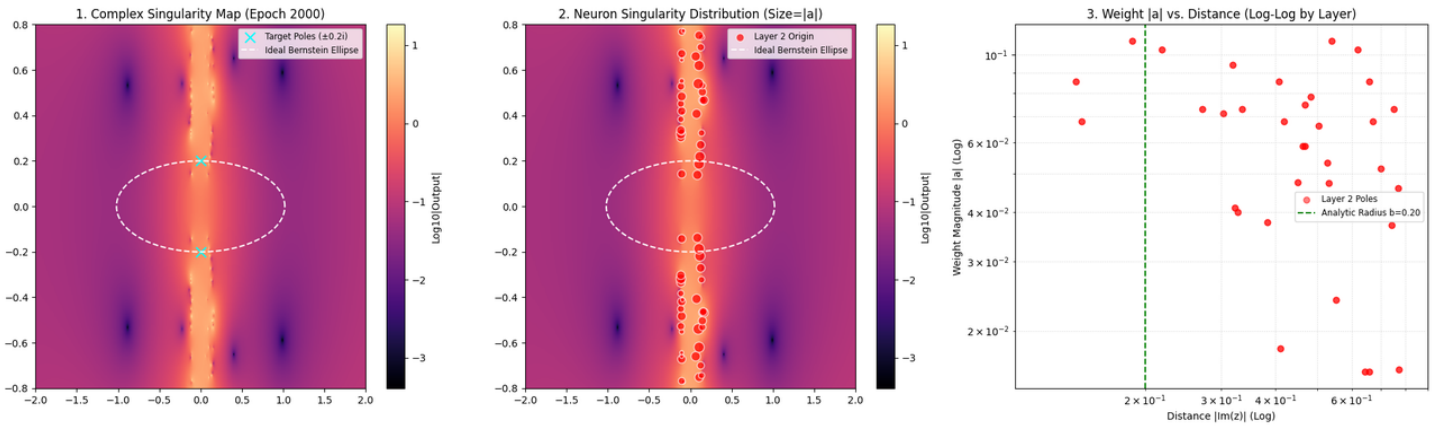
colab链接：
https://colab.research.google.com/drive/1OhBTnSRj19DH8QpSx0T0LFtpJ5q5OrbA?usp=sharing

实验结论：不符合预期，权重系数的大小和重要性分布一样

实验目的

仍然想在权重系数的视角下验证梯度下降优化是在让网络伯恩斯坦椭圆靠近目标伯恩斯坦椭圆，即目标伯恩斯坦椭圆区域上的权重系数要比内外的权重系数要大。 实验记录_20260129_双隐藏层mlp中奇点权重系数变化 已经做了这部分探究，但是没有达到预期，所以换一种方式进行探究。

通过将奇点的权重系数置为0，观察预测输出的前后变化，画出前后变化的二范数和距离的关系图，预期在目标解析区域附近的扰动大，往内走往外走都会变小

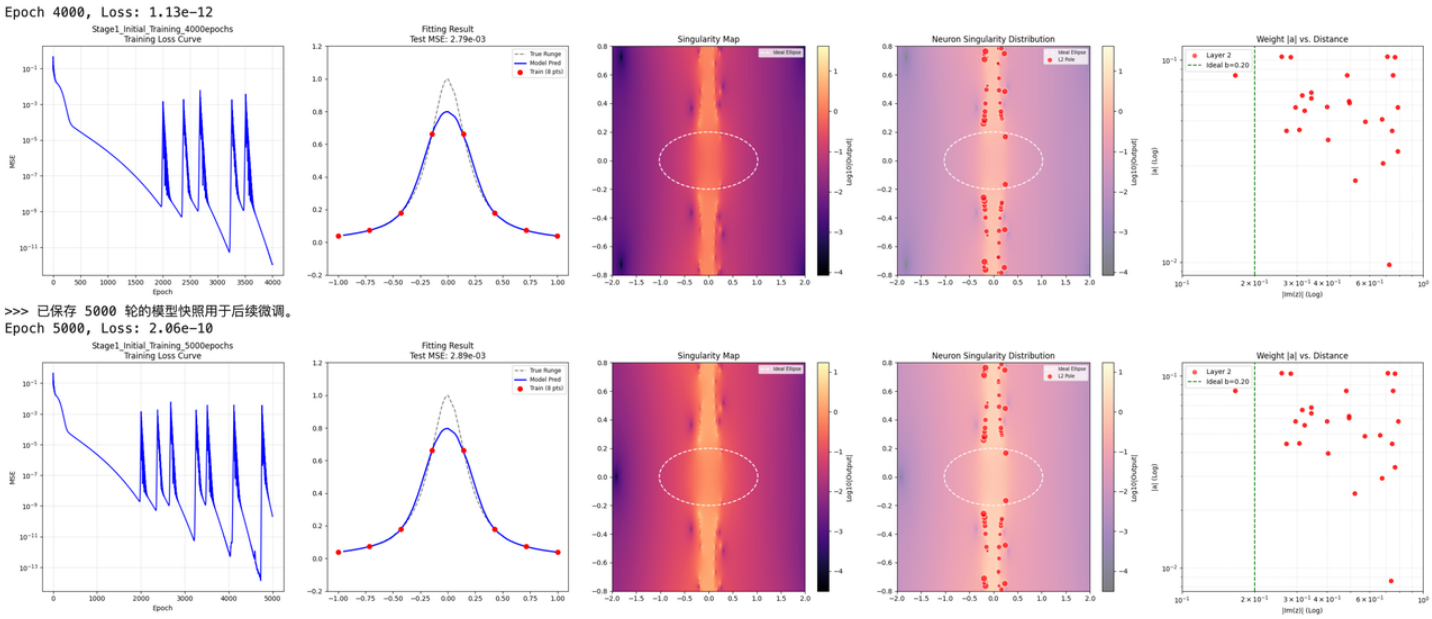


实验步骤

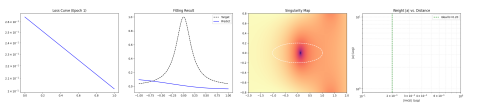
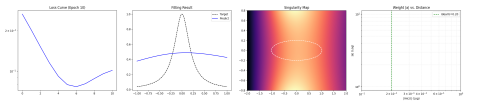
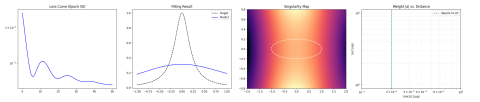
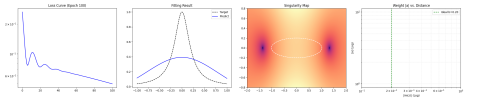
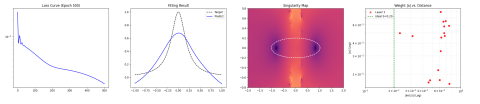
初始训练

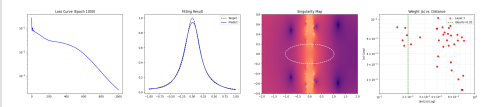
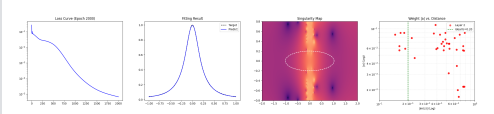
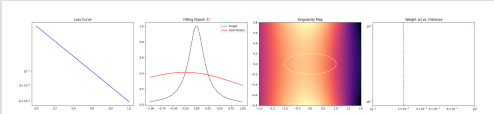
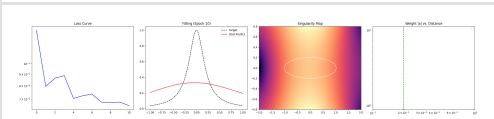
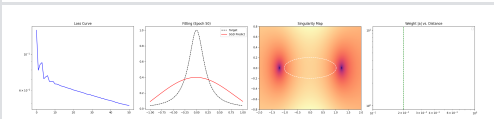
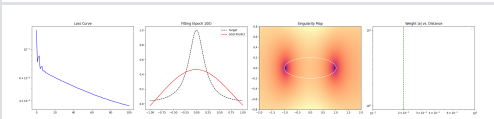
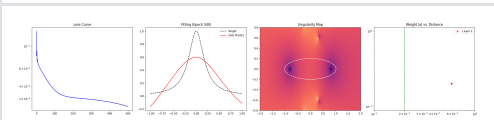
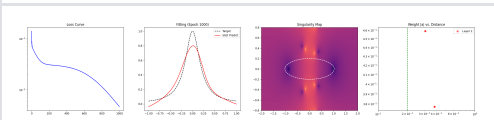
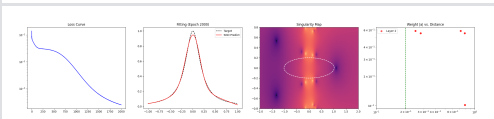
- 模型和函数：构建双层mlp，目标函数为beta0.2的龙格函数，这样目标解析区域会大于网络解析区域
- 训练集：并在【-1，1】上均匀采样2000个点
- 训练过程：adam优化器，学习率1e-4，训练轮次 [1000, 2000, 4000, 5000]

- 每个训练完后得到loss图，拟合图和奇点图（下图的图一图二图四）。
- 然后遍历所有激活函数前的权重系数，并每次将一个权重系数置为0，计算在这种情况下的拟合输出和**该激活函数对应的奇点位置**，画出三个图
- 1.拟合输出的前后变化情况（虚线为前输出，蓝线为后输出，标题要标注好变化的二范数和奇点位置）
- 2.改变后的奇点图上（对应位置加一个白色点）
- 3.二范数 vs 奇点位置log图



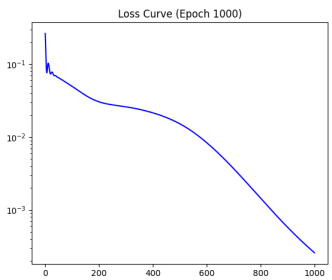
实验组

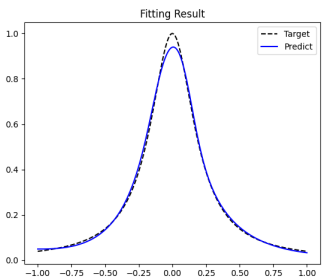
序号	epoch	layers	神经元个数	学习率	优化器	实验结果	实验图
1	1	2	256	1e-4	adam	和之前的权重系数分布一样	
	10						
	50						
	100						
	500						

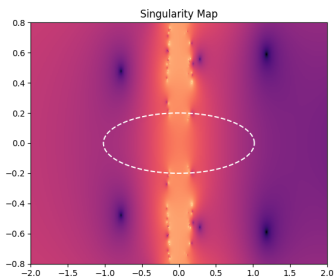
	1000						
	2000						
2	1	2	256	1e-4	sgd	sgd优化器下下，奇点没有进到伯恩斯坦椭圆内的	
	10						
	50						
	100						
	500						
	1000						
	2000						

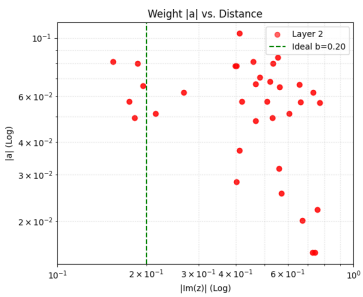
实验图

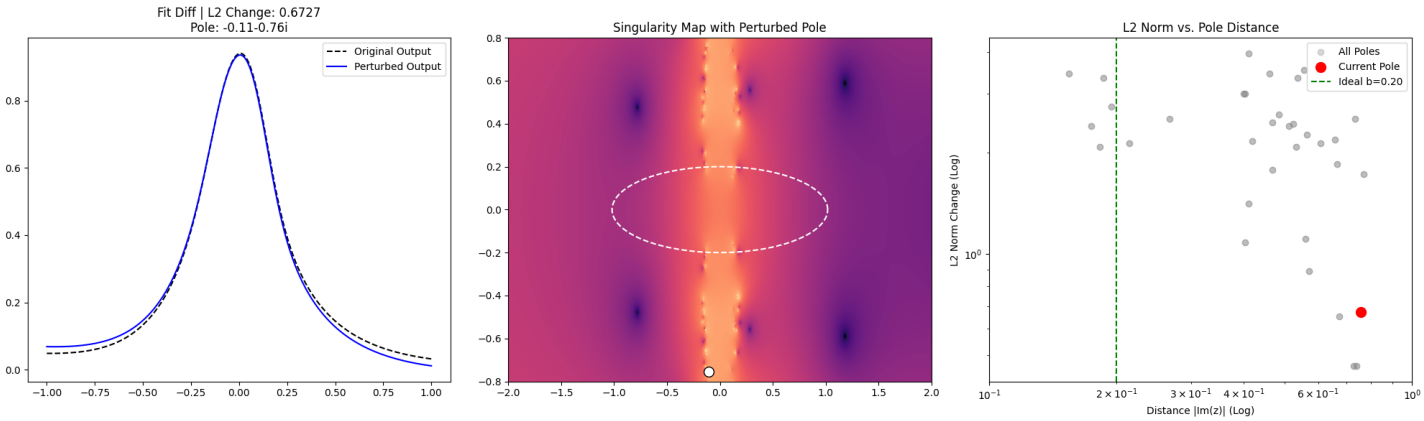
Adam

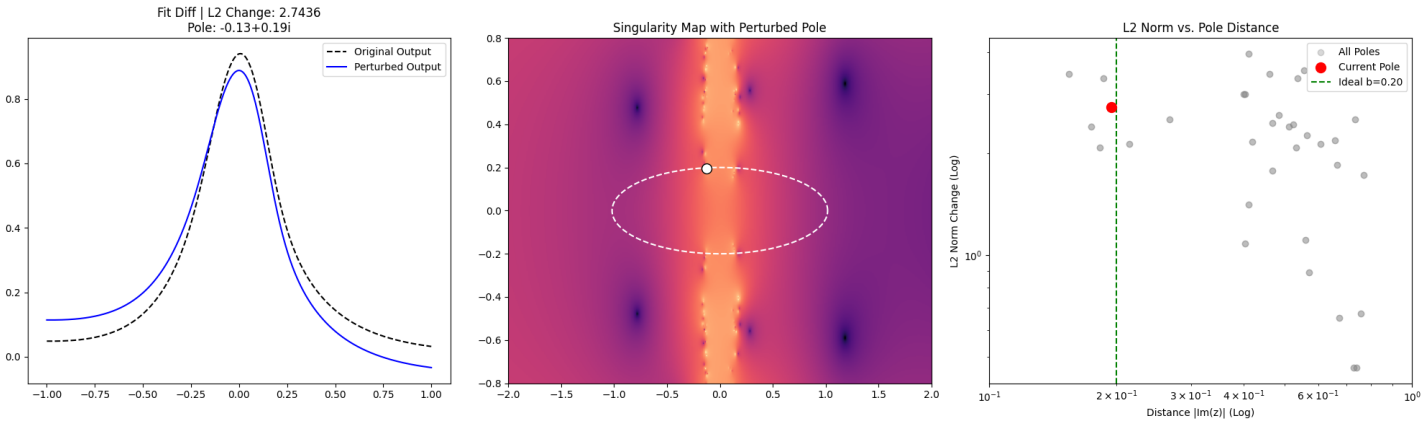


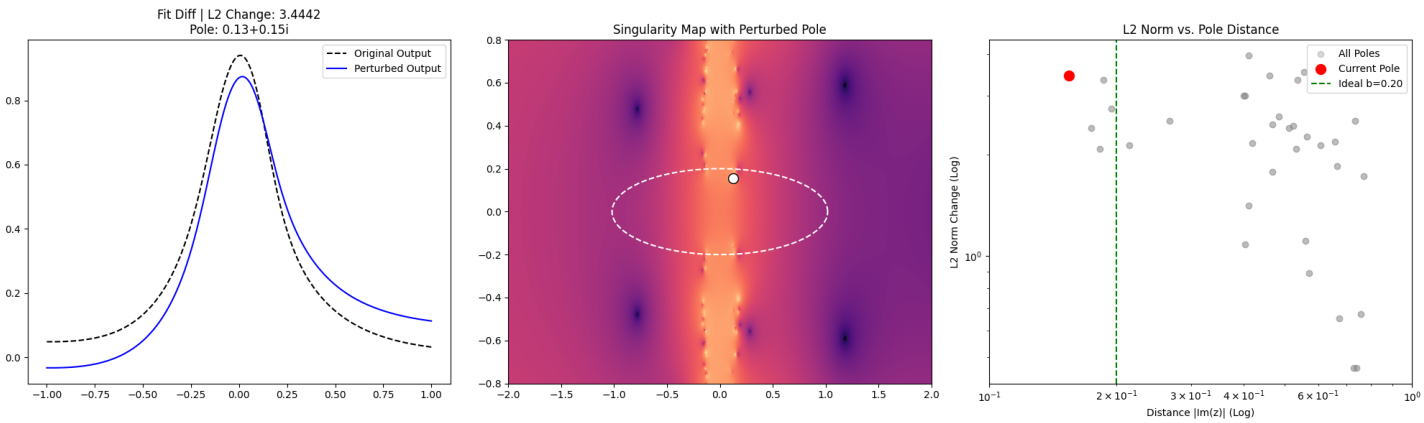


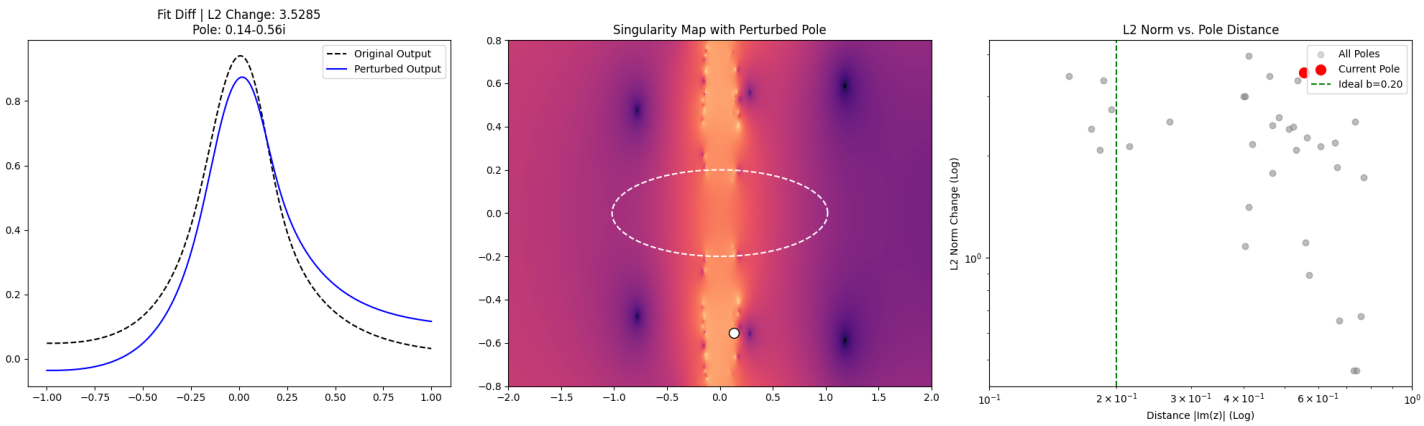




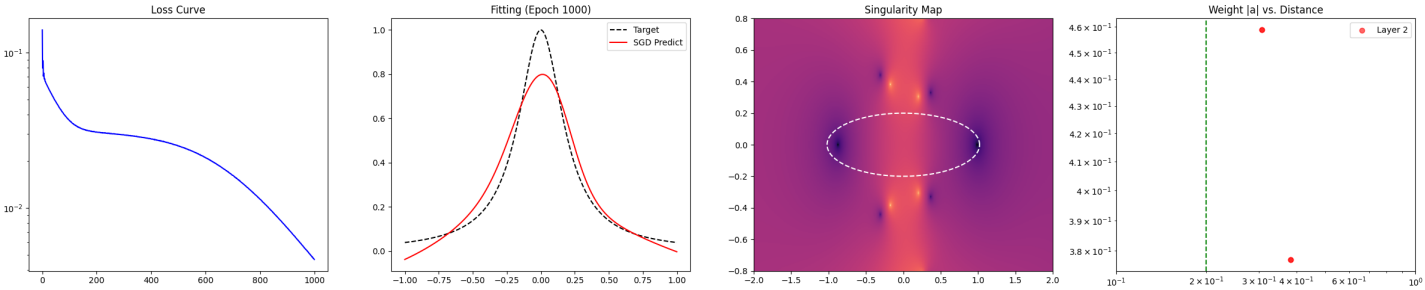

Fit Diff | L2 Change: 0.6727
Pole: -0.11-0.76i
Original Output
Perturbed Output
Singularity Map with Perturbed Pole
L2 Norm vs. Pole Distance
All Poles
Current Pole
Ideal b=0.20
L2 Norm Change (Log)
Distance |Im(z)| (Log)

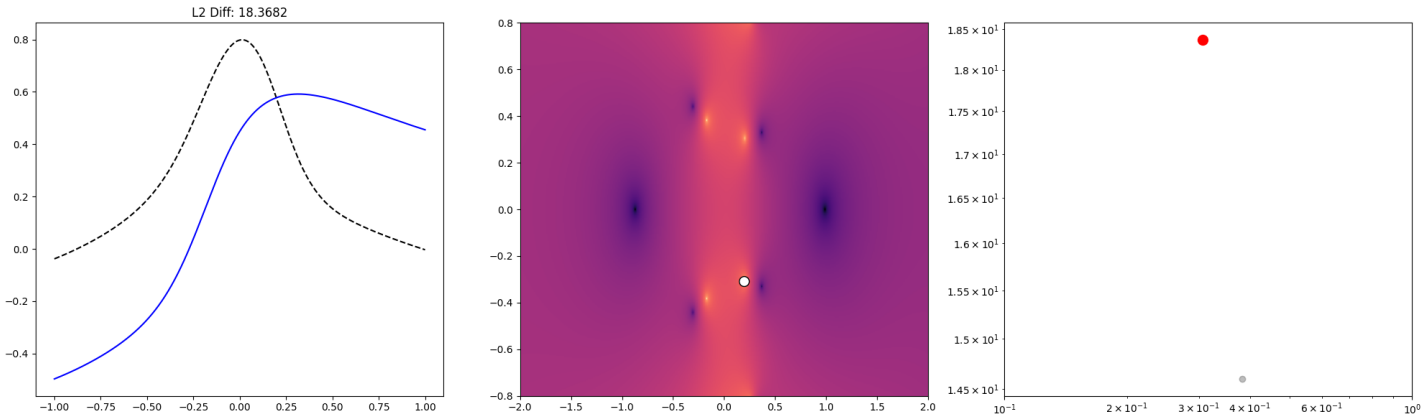

Fit Diff | L2 Change: 2.7436
Pole: -0.13+0.19i
Original Output
Perturbed Output
Singularity Map with Perturbed Pole
L2 Norm vs. Pole Distance
All Poles
Current Pole
Ideal b=0.20
L2 Norm Change (Log)
Distance |Im(z)| (Log)


Fit Diff | L2 Change: 3.4442
Pole: 0.13+0.15i
Original Output
Perturbed Output
Singularity Map with Perturbed Pole
L2 Norm vs. Pole Distance
All Poles
Current Pole
Ideal b=0.20
L2 Norm Change (Log)
Distance |Im(z)| (Log)


Fit Diff | L2 Change: 3.5285
Pole: 0.14-0.56i
Original Output
Perturbed Output
Singularity Map with Perturbed Pole
L2 Norm vs. Pole Distance
All Poles
Current Pole
Ideal b=0.20
L2 Norm Change (Log)
Distance |Im(z)| (Log)

SGD


Loss Curve
Fitting (Epoch 1000)
Target
SGD Predict
Singularity Map
Weight |a| vs. Distance
Layer 2


L2 Diff: 18.3682